

MA4110/MA5020: Computational Methods for Fluid Flow

Lecture 10: Introduction to the Finite Volume Method, Control Volumes, and Godunov's Scheme

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1 Introduction: Why Do We Need a New Method?

In Lectures 6 through 9, we built the **Finite Difference Method (FDM)**. The whole idea of FDM is simple: track the value of u at a point x_j , and update it in time using derivatives $\frac{\partial u}{\partial t}$ and $\frac{\partial u}{\partial x}$ approximated from Taylor series. In Lecture 9 we also learned one important rule for FDM: the scheme must be written in **conservative flux form**, otherwise it computes the wrong shock speed.

But there is a deeper problem that conservative form alone cannot fix. Ask a very basic question: *what is $\frac{\partial u}{\partial x}$ right at a shock?* A shock is a jump: u changes almost instantly from one value to another. A derivative measures how steep a graph is, but a vertical jump has infinite steepness. So $\frac{\partial u}{\partial x} \rightarrow \pm\infty$ at the shock, and the Taylor series that FDM is built on simply does not make sense there.

- **FDM's job is to track point values** $u(x_j, t)$ using derivatives.
- **But derivatives do not exist at a jump.** So a method built entirely on derivatives has nothing meaningful to compute exactly where a shock sits.

The fix is simple. Instead of asking "what is u at this exact point?", ask "how much of the conserved quantity (mass, momentum, energy) is contained in this small box around the point?" A box can hold a perfectly well-defined, finite amount of material even if u jumps somewhere inside it. Averaging over the box simply smooths the jump away.

This is the entire idea of the **Finite Volume Method (FVM)**. Instead of tracking the point value $u(x_j, t)$ at a grid node, FVM tracks the **cell average** of u inside a small box (control volume) surrounding that point:

$$\bar{u}_i(t) \equiv \frac{1}{\Delta x_i} \int_{x_{i-1/2}}^{x_{i+1/2}} u(x, t) dx \quad (1)$$

Here $\bar{u}_i(t)$ is just the average value of u inside the box $I_i = [x_{i-1/2}, x_{i+1/2}]$. Averaging a jump always gives an ordinary, finite number, so cell averages stay perfectly well-behaved even when a shock passes right through the box.

In this lecture, we begin **Module 3: Finite Volume Methods**. In plain terms, here is our roadmap:

1. Write down the exact conservation law for a single box in space and time (no approximation yet).
2. Show that this box-balance equation is **exact**: the only thing left to approximate is the flux flowing through the box's two edges.
3. Work out, by hand, what happens when two boxes with different values meet at an edge (the **Riemann problem**), first for simple advection, then for Burgers' equation.
4. Use that result to build **Godunov's method (1959)**, the first scheme that gets shocks exactly right.
5. Check the properties that make Godunov's scheme so reliable: it never overshoots and never creates fake wiggles.

2 The 1D Control Volume Formulation

2.1 Space-Time Integration of the Conservation Law

Consider the scalar one-dimensional hyperbolic conservation law:

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0 \quad (2)$$

Divide the spatial domain into disjoint computational cells (control volumes) $I_i = [x_{i-1/2}, x_{i+1/2}]$ of width $\Delta x_i = x_{i+1/2} - x_{i-1/2}$, with cell centers $x_i = \frac{1}{2}(x_{i-1/2} + x_{i+1/2})$. The time domain is discretized into time intervals $[t^n, t^{n+1}]$ with step size $\Delta t = t^{n+1} - t^n$.

Now integrate equation (2) over the two-dimensional **space-time control volume**:

$$\begin{aligned} \Omega_i^n &\equiv [x_{i-1/2}, x_{i+1/2}] \times [t^n, t^{n+1}] \\ \iint_{\Omega_i^n} \left(\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} \right) dx dt &= 0 \end{aligned} \quad (3)$$

Splitting the double integral into spatial and temporal components:

$$\int_{x_{i-1/2}}^{x_{i+1/2}} \left(\int_{t^n}^{t^{n+1}} \frac{\partial u}{\partial t} dt \right) dx + \int_{t^n}^{t^{n+1}} \left(\int_{x_{i-1/2}}^{x_{i+1/2}} \frac{\partial f(u)}{\partial x} dx \right) dt = 0 \quad (4)$$

Figure 1 shows this space-time box explicitly. The shaded rectangle is Ω_i^n : its bottom edge carries the known cell average $\bar{u}_i^n$ at time t^n , and its top edge carries the unknown cell average $\bar{u}_i^{n+1}$ we want to compute at time t^{n+1} . Nothing happens through the top and bottom edges themselves (there is no flux *in time*); the entire change in the box's content comes from material crossing its two *vertical* edges, i.e., the fluxes $F_{i-1/2}^n$ and $F_{i+1/2}^n$ at the interfaces $x_{i-1/2}$ and $x_{i+1/2}$.

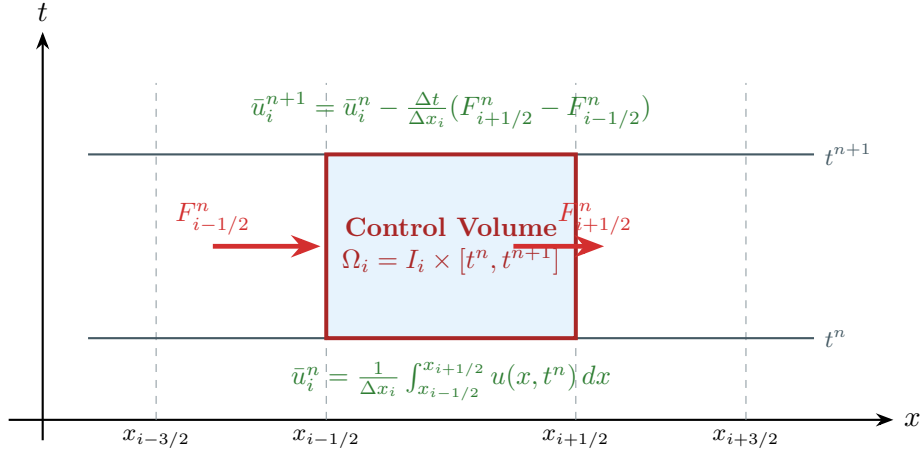


Figure 1: Space-time control volume $\Omega_i = [x_{i-1/2}, x_{i+1/2}] \times [t^n, t^{n+1}]$. Cell averages $\bar{u}_i^n$ and $\bar{u}_i^{n+1}$ represent the state integrated along the bottom and top boundaries. Changes in cell average are driven exclusively by the net time-integrated fluxes $F_{i-1/2}^n$ and $F_{i+1/2}^n$ crossing the left and right cell interfaces.

Applying the Fundamental Theorem of Calculus to both inner integrals:

$$\int_{x_{i-1/2}}^{x_{i+1/2}} [u(x, t^{n+1}) - u(x, t^n)] dx + \int_{t^n}^{t^{n+1}} [f(u(x_{i+1/2}, t)) - f(u(x_{i-1/2}, t))] dt = 0 \quad (5)$$

Rearranging terms so that the state at the new time level t^{n+1} is isolated:

$$\int_{x_{i-1/2}}^{x_{i+1/2}} u(x, t^{n+1}) dx = \int_{x_{i-1/2}}^{x_{i+1/2}} u(x, t^n) dx - \left[\int_{t^n}^{t^{n+1}} f(u(x_{i+1/2}, t)) dt - \int_{t^n}^{t^{n+1}} f(u(x_{i-1/2}, t)) dt \right] \quad (6)$$

2.2 The Exact Finite Volume Update Equation

Using the definition of the cell average $\bar{u}_i$ from equation (1), we define the **exact time-averaged numerical flux** through cell interface $x_{i+1/2}$ over the time interval $[t^n, t^{n+1}]$:

$$F_{i+1/2}^n \equiv \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f(u(x_{i+1/2}, t)) dt \quad (7)$$

Dividing equation (6) by the cell length Δx_i yields the foundational update formula of the Finite Volume Method:

$$\bar{u}_i^{n+1} = \bar{u}_i^n - \frac{\Delta t}{\Delta x_i} (F_{i+1/2}^n - F_{i-1/2}^n) \quad (8)$$

The Central Miracle of Finite Volume Methods

Equation (8) is **mathematically exact**. No Taylor series truncation, spatial differencing, or temporal discretization errors have been introduced!

If one could evaluate the exact time-averaged boundary fluxes $F_{i\pm 1/2}^n$, equation (8) would yield the *exact* cell averages of the true physical solution for all time.

Therefore, in the Finite Volume framework, **all numerical modeling is concentrated into a single task**: approximating the interface flux $F_{i+1/2}^n$ given only the discrete cell averages $\{\bar{u}_i^n\}$.

2.3 Discrete Global Conservation

Summing equation (8) multiplied by Δx_i over all cells in a domain spanning $i = 1, 2, \dots, N$:

$$\sum_{i=1}^N \bar{u}_i^{n+1} \Delta x_i = \sum_{i=1}^N \bar{u}_i^n \Delta x_i - \Delta t \sum_{i=1}^N (F_{i+1/2}^n - F_{i-1/2}^n) \quad (9)$$

Because the numerical flux leaving cell i through its right face ($F_{i+1/2}^n$) is identically equal to the flux entering cell $i+1$ through its left face, every interior flux appears twice in the sum with opposite signs and cancels; only the two fluxes at the outer physical boundaries survive. In other words, the interior summation telescopes to zero:

$$\sum_{i=1}^N (F_{i+1/2}^n - F_{i-1/2}^n) = F_{N+1/2}^n - F_{1/2}^n \quad (10)$$

Hence, the total discrete mass inside the domain changes exclusively through the physical boundaries:

$$\sum_{i=1}^N \bar{u}_i^{n+1} \Delta x_i = \sum_{i=1}^N \bar{u}_i^n \Delta x_i - \Delta t (F_{N+1/2}^n - F_{1/2}^n) \quad (11)$$

Exact conservation is guaranteed on any grid (uniform or non-uniform) and for any flow (smooth or containing shocks)!

3 The Interface Dilemma and the Riemann Problem

3.1 The Dilemma at the Interface

To evaluate the flux integral (7), we must know the state $u(x_{i+1/2}, t)$ along the vertical interface line $x = x_{i+1/2}$ for $t \in [t^n, t^{n+1}]$.

However, the computational data at time t^n consists only of cell averages $\{\bar{u}_i^n\}$. If we represent the flow inside each cell by its cell average (piecewise constant):

$$u(x, t^n) = \bar{u}_i^n \quad \text{for } x \in [x_{i-1/2}, x_{i+1/2}] \quad (12)$$

then at the interface $x_{i+1/2}$, the state is **discontinuous**:

$$u(x_{i+1/2}^-, t^n) = \bar{u}_i^n \equiv u_L, \quad u(x_{i+1/2}^+, t^n) = \bar{u}_{i+1}^n \equiv u_R \quad (13)$$

What is the state $u(x_{i+1/2}, t)$ for $t > t^n$? Does the left state u_L prevail, or the right state u_R , or does an entirely new intermediate state form?

3.2 Definition of the Riemann Problem

The mathematical problem of determining how an isolated jump discontinuity evolves in time under a conservation law is called the **Riemann Problem** (first studied by Bernhard Riemann in 1860 for gas dynamics):

The Riemann Problem for a Scalar Conservation Law

The Riemann problem is the initial value problem (IVP) for a conservation law subject to piecewise-constant initial data with a single jump:

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0, \quad u(x, 0) = \begin{cases} u_L, & x < 0 \\ u_R, & x > 0 \end{cases} \quad (14)$$

Because the differential equation and the initial data contain no intrinsic length scale or time scale, the solution is **self-similar**: it depends only on the similarity ratio $\xi = x/t$:

$$u(x, t) = W\left(\frac{x}{t}\right) \quad (15)$$

Any feature (shock or characteristic) emanates from the origin ($x = 0, t = 0$) along a straight ray in the (x, t) plane.

We will repeatedly need the value of this solution exactly along the stationary interface line $x = 0$ (i.e., at $\xi = 0$). We give it its own shorthand:

$$u^* \equiv W(0) = u(x = 0, t) \quad \text{for any } t > 0 \quad (16)$$

In words: u^* is simply "the state that sits on the cell interface," and since the solution is self-similar, this one number is the same for every $t > 0$.

3.3 Riemann Solution for Linear Advection

For linear advection $f(u) = au$ ($a \in \mathbb{R}$):

$$u_t + au_x = 0 \quad (17)$$

The jump simply propagates along the characteristic ray $x = at$ without changing shape:

$$u(x, t) = u_0(x - at) = \begin{cases} u_L, & x - at < 0 \iff \frac{x}{t} < a \\ u_R, & x - at > 0 \iff \frac{x}{t} > a \end{cases} \quad (18)$$

Using the shorthand from equation (16), the interface state is:

$$u^* = \begin{cases} u_L, & \text{if } a > 0 \\ u_R, & \text{if } a < 0 \end{cases} \quad (19)$$

This confirms that the physical upwind state is the exact solution of the linear Riemann problem along the cell boundary!

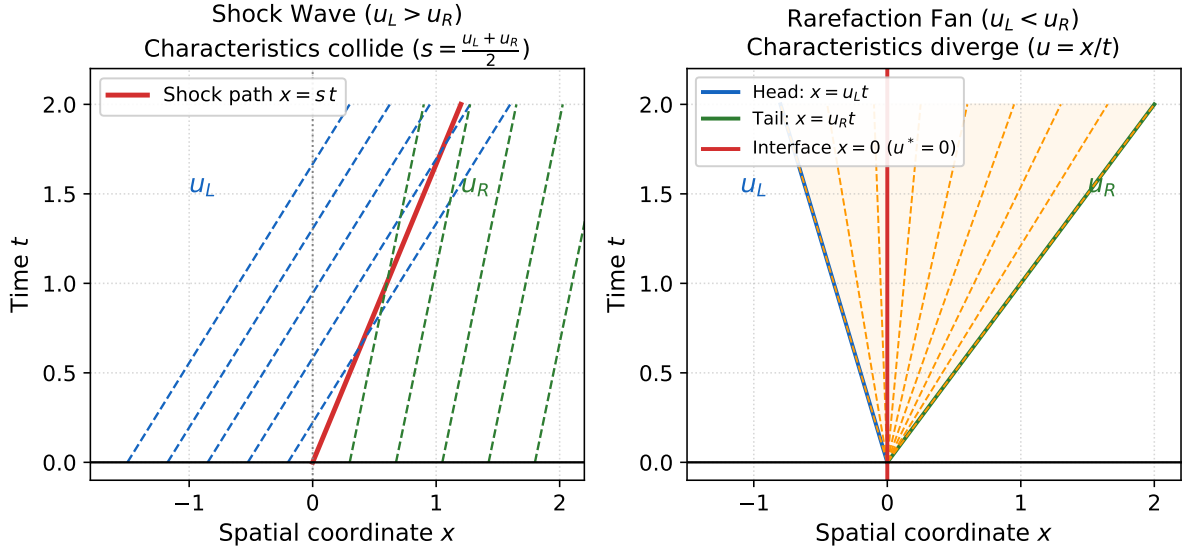


Figure 2: Structure of the Riemann problem in the (x, t) plane for a convex flux ($f''(u) > 0$). Left: Shock wave ($u_L > u_R$), characteristics converge into a shock moving at speed s . Right: Rarefaction fan ($u_L < u_R$), characteristics diverge and create a continuous expansion fan bounded by $x = f'(u_L)t$ and $x = f'(u_R)t$.

3.4 Riemann Solution for Convex Scalar Conservation Laws (Burgers' Equation)

For a nonlinear conservation law with strictly convex flux ($f''(u) > 0$, e.g., Burgers' equation $f(u) = \frac{1}{2}u^2$), the solution depends on the relative ordering of u_L and u_R . Figure 2 previews both possibilities before we derive them in detail below: a shock when $u_L > u_R$ (left panel), or a rarefaction fan when $u_L < u_R$ (right panel).

3.4.1 Case 1: Shock Wave ($u_L > u_R$)

When $u_L > u_R$, characteristic lines from the left (speed $f'(u_L)$) travel faster than characteristic lines from the right (speed $f'(u_R)$) and collide. The unique weak solution satisfying the Lax entropy condition (Lecture 5) is a **discontinuous shock wave** propagating at speed s given by the Rankine-Hugoniot condition (Lecture 5):

$$s = \frac{f(u_L) - f(u_R)}{u_L - u_R} \quad (20)$$

For Burgers' equation, $s = \frac{1}{2}(u_L + u_R)$. The solution is:

$$u(x, t) = \begin{cases} u_L, & x < st \\ u_R, & x > st \end{cases} \quad (21)$$

Along the interface $x = 0$, the state is:

$$u^* = \begin{cases} u_L, & \text{if } s > 0 \\ u_R, & \text{if } s < 0 \end{cases} \quad (22)$$

3.4.2 Case 2: Rarefaction Wave ($u_L < u_R$)

When $u_L < u_R$, characteristics diverge and leave an empty region between $x = f'(u_L)t$ and $x = f'(u_R)t$. The entropy-satisfying weak solution is a smooth, continuous **rarefaction fan** (expansion fan).

Substituting the similarity ansatz $u(x, t) = W(\xi)$ with $\xi = x/t$ into equation (2):

$$\frac{\partial u}{\partial t} + f'(u) \frac{\partial u}{\partial x} = -\frac{\xi}{t} W'(\xi) + \frac{1}{t} f'(W(\xi)) W'(\xi) = \frac{W'(\xi)}{t} [f'(W(\xi)) - \xi] = 0 \quad (23)$$

Inside the fan, $W'(\xi) \neq 0$, which yields the characteristic speed equation:

$$f'(W(\xi)) = \xi \implies W(\xi) = (f')^{-1}(\xi) \quad (24)$$

For Burgers' equation, $f'(u) = u$, so the inverse function is simply $(f')^{-1}(\xi) = \xi = x/t$. Thus, the complete rarefaction solution is:

$$u(x, t) = \begin{cases} u_L, & \frac{x}{t} \leq u_L \\ \frac{x}{t}, & u_L < \frac{x}{t} < u_R \\ u_R, & \frac{x}{t} \geq u_R \end{cases} \quad (25)$$

The Transonic Rarefaction and the Sonic Point. When $u_L < 0 < u_R$, the expansion fan straddles the interface line $x/t = 0$. The interface state is evaluated directly by setting $\xi = 0$ in equation (25):

$$u^* = 0 \quad (\text{the sonic point where } f'(u) = 0) \quad (26)$$

The flux at the interface is therefore $f(u^*) = f(0) = 0$. This special situation is called a **transonic rarefaction**: on the left side of the interface, characteristics move left ($f'(u_L) < 0$, since $u_L < 0$); on the right side, characteristics move right ($f'(u_R) > 0$, since $u_R > 0$). So information moves away from the interface in *both* directions at once, and the interface itself sits exactly at the one point in the fan where the characteristic speed is zero.

4 Godunov's Foundational Method (1959)

In 1959, the Soviet mathematician Sergei K. Godunov proposed a breakthrough methodology: *use the exact solution of local Riemann problems to compute the numerical fluxes of a finite volume scheme.*

4.1 The Three Steps of Godunov's Scheme

Given the cell averages $\{\bar{u}_i^n\}$ at time level t^n :

Step 1. Reconstruction: Construct a piecewise-constant function $\tilde{u}(x, t^n)$ by setting the state in each cell I_i equal to its cell average:

$$\tilde{u}(x, t^n) = \bar{u}_i^n \quad \text{for } x \in [x_{i-1/2}, x_{i+1/2}] \quad (27)$$

Step 2. Evolution: Solve the exact Riemann problem at each cell interface $x_{i+1/2}$ with left state $u_L = \bar{u}_i^n$ and right state $u_R = \bar{u}_{i+1}^n$. This generates waves propagating outward from each interface.

Step 3. Cell Averaging (Flux Evaluation): No disturbance from an interface's Riemann problem (shock or rarefaction fan) can travel faster than the fastest speed present in the solution, $\max |f'(u)|$. So in a time Δt , the region it could possibly affect, its **wave cone**, extends at most $\max |f'(u)| \cdot \Delta t$ to the left or right of its interface. Two neighbouring interfaces, say $x_{i-1/2}$ and $x_{i+1/2}$, are a distance Δx apart, i.e., $\Delta x/2$ on either side of the cell center. So as long as each wave cone spreads by at most $\Delta x/2$ before time t^{n+1} , the two cones cannot yet have reached each other:

$$\max |f'(u)| \cdot \Delta t \leq \frac{\Delta x}{2} \implies \Delta t \leq \frac{\Delta x}{2 \max |f'(u)|} \quad (28)$$

This is exactly the **CFL condition** we impose. Figure 3 (left panel) shows it directly: under this bound, the wave cones from $x_{i-1/2}$ and $x_{i+1/2}$ stay inside their own halves of the cell and never

touch by t^{n+1} . That is why we are allowed to say the state along the vertical line $x = x_{i+1/2}$ stays **constant in time**: nothing happening at the neighbouring interfaces has had time to reach $x_{i+1/2}$ yet, so the value there depends only on the single Riemann problem centered at $x_{i+1/2}$.

What if Δt is a little larger, so that $\frac{1}{2} < \nu \leq 1$? Figure 3 (right panel) shows this case: the two wave cones now overlap, so the interface state is no longer exactly constant in time. Even so, the Godunov update (8) remains exact, because it only ever needs the *net* flux through each interface, and conservation still fixes that net flux correctly even when neighbouring wave cones overlap. So $\nu \leq 1/2$ is only needed for the simple constant-in-time argument used above, while the sharp requirement for monotonicity (Section 5.1) is the looser bound $\nu \leq 1$.

To write this precisely, we need one more piece of notation. Godunov's method solves a *different* Riemann problem at every interface (each with its own u_L, u_R), so we write $u_{\text{RP}}(\xi; u_L, u_R)$ for “the self-similar solution $W(\xi)$ of the Riemann problem with left state u_L and right state u_R ”: this is exactly the $u^* \equiv W(0)$ from equation (16), just with the states written out explicitly so we can tell interfaces apart. Using this notation, the interface state at $x_{i+1/2}$ is:

$$u(x_{i+1/2}, t) = u_{\text{RP}}(0; \bar{u}_i^n, \bar{u}_{i+1}^n) \equiv u_{i+1/2}^* \quad \text{for all } t \in [t^n, t^{n+1}] \quad (29)$$

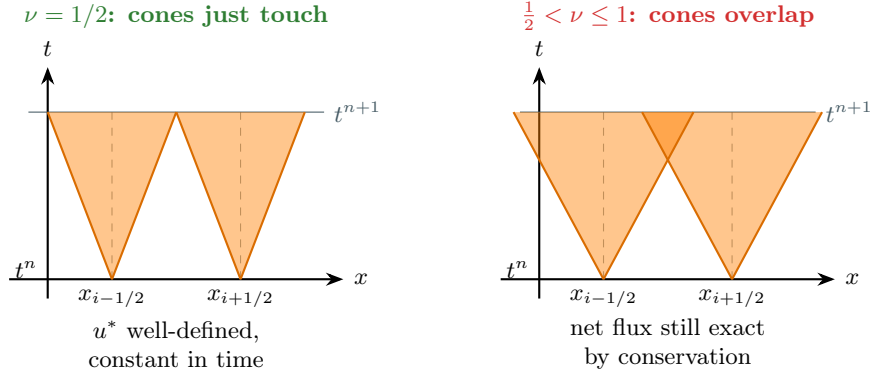


Figure 3: Wave cones from the two interfaces of a cell. Left: $\nu = 1/2$, the cones just touch. Right: $\frac{1}{2} < \nu \leq 1$, the cones overlap (darker shaded region).

Because the interface state $u_{i+1/2}^*$ is constant over the entire time step, the exact time-integral in the flux definition (7) simplifies trivially:

$$F_{i+1/2}^{\text{Godunov}} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f(u_{i+1/2}^*) dt = f(u_{i+1/2}^*) \quad (30)$$

Definition of the Godunov Numerical Flux

The **Godunov numerical flux** is the physical flux evaluated at the self-similar interface state of the exact Riemann problem:

$$F_{i+1/2}^{\text{Godunov}} = f(u_{\text{RP}}(0; \bar{u}_i^n, \bar{u}_{i+1}^n)) \quad (31)$$

4.2 General Closed-Form Godunov Flux for Convex Fluxes

For any scalar conservation law with strictly convex flux ($f''(u) > 0$), Osher (1984) showed that Godunov's flux can be written compactly as:

$$F^{\text{Godunov}}(u_L, u_R) = \begin{cases} \min_{u \in [u_L, u_R]} f(u), & \text{if } u_L \leq u_R \quad (\text{rarefaction}) \\ \max_{u \in [u_R, u_L]} f(u), & \text{if } u_L > u_R \quad (\text{shock}) \end{cases} \quad (32)$$

4.3 Explicit Formulas for Common Conservation Laws

4.3.1 Linear Advection ($f(u) = au$)

For $f(u) = au$, the min/max formula (32) yields:

$$F^{\text{Godunov}}(u_L, u_R) = \begin{cases} au_L, & a > 0 \\ au_R, & a < 0 \end{cases} = \frac{a}{2}(u_L + u_R) - \frac{|a|}{2}(u_R - u_L) \quad (33)$$

Substituting this flux into the FVM update (8) recovers the **classical first-order upwind scheme!** This proves that upwinding is not an ad-hoc trick: it is the exact Finite Volume discretization of linear wave propagation!

4.3.2 Inviscid Burgers' Equation ($f(u) = \frac{1}{2}u^2$)

For $f(u) = \frac{1}{2}u^2$, the sonic point occurs at $u_s = 0$ where $f'(0) = 0$. Evaluating formula (32) leads to the explicit five-branch decision rule:

$$F^{\text{Godunov}}(u_L, u_R) = \begin{cases} \frac{1}{2}u_L^2, & u_L \geq u_R \text{ and } s = \frac{u_L + u_R}{2} > 0 \quad (\text{right-moving shock}) \\ \frac{1}{2}u_R^2, & u_L \geq u_R \text{ and } s = \frac{u_L + u_R}{2} \leq 0 \quad (\text{left-moving shock}) \\ \frac{1}{2}u_L^2, & u_L < u_R \text{ and } u_L \geq 0 \quad (\text{supersonic right rarefaction}) \\ \frac{1}{2}u_R^2, & u_L < u_R \text{ and } u_R \leq 0 \quad (\text{supersonic left rarefaction}) \\ 0, & u_L < 0 < u_R \quad (\text{transonic rarefaction: sonic point } u^* = 0) \end{cases} \quad (34)$$

5 Mathematical Properties of Godunov's Scheme

5.1 Monotonicity and Godunov's Scheme

A finite difference or finite volume scheme expressed in the operator form:

$$\bar{u}_i^{n+1} = \mathcal{H}(\bar{u}_{i-1}^n, \bar{u}_i^n, \bar{u}_{i+1}^n) \quad (35)$$

is defined to be **monotone** if $\mathcal{H}$ is a non-decreasing function of each of its arguments:

$$\boxed{\frac{\partial \mathcal{H}}{\partial \bar{u}_{i-1}} \geq 0, \quad \frac{\partial \mathcal{H}}{\partial \bar{u}_i} \geq 0, \quad \frac{\partial \mathcal{H}}{\partial \bar{u}_{i+1}} \geq 0} \quad (36)$$

Theorem: Monotonicity of Godunov's Scheme

Under the standard CFL condition:

$$\nu \equiv \max |f'(u)| \frac{\Delta t}{\Delta x} \leq 1$$

Godunov's scheme is a **monotone scheme**.

Proof Sketch for Linear Advection. For $f(u) = au$ with $a > 0$, the Godunov scheme is:

$$\bar{u}_i^{n+1} = \mathcal{H}(\bar{u}_{i-1}^n, \bar{u}_i^n) = \bar{u}_i^n - \nu(\bar{u}_i^n - \bar{u}_{i-1}^n) = \nu \bar{u}_{i-1}^n + (1 - \nu) \bar{u}_i^n$$

Taking partial derivatives:

$$\frac{\partial \mathcal{H}}{\partial \bar{u}_{i-1}} = \nu \geq 0, \quad \frac{\partial \mathcal{H}}{\partial \bar{u}_i} = 1 - \nu \geq 0 \iff \nu \leq 1, \quad \frac{\partial \mathcal{H}}{\partial \bar{u}_{i+1}} = 0 \geq 0$$

All partial derivatives are non-negative if and only if $\nu \leq 1$. □

5.2 The TVD (Total Variation Diminishing) Property

The **Total Variation** of a discrete grid function $\bar{\mathbf{u}} \equiv \{\bar{u}_i\}_i$ (the entire collection of cell averages at one time level) is defined as:

$$\text{TV}(\bar{\mathbf{u}}) \equiv \sum_{i=-\infty}^{\infty} |\bar{u}_{i+1} - \bar{u}_i| \quad (37)$$

A scheme is **Total Variation Diminishing (TVD)** if:

$$\text{TV}(\bar{\mathbf{u}}^{n+1}) \leq \text{TV}(\bar{\mathbf{u}}^n) \quad (38)$$

A fundamental lemma of Harten (1983) establishes:

Every monotone scheme is Total Variation Diminishing (TVD).

Consequences of the TVD property:

- **No new local extrema:** The scheme cannot create spurious wiggles, ripples, or overshoots near discontinuities ($u_{\min}^n \leq \bar{u}_i^{n+1} \leq u_{\max}^n$).
- **Monotonicity preservation:** If the initial data $\bar{\mathbf{u}}^0$ is monotonically decreasing (such as a shock profile), the numerical solution remains monotonically decreasing for all time steps $n > 0$.

5.3 Order of Accuracy and First-Order Numerical Diffusion

Why does Godunov's scheme not violate Godunov's Barrier Theorem (Lecture 8)? Because **Godunov's scheme is strictly first-order accurate** ($\mathcal{O}(\Delta x, \Delta t)$).

The first-order accuracy stems directly from **Step 1 (Piecewise Constant Reconstruction)**. The cell averages themselves are not the problem: the flat block in cell I_i has exactly the average $\bar{u}_i$, so no conserved quantity is lost. What is lost is the *shape* of the solution inside the cell. Taylor-expanding a smooth $u(x)$ about the cell centre x_i , the **projection error** of the flat reconstruction $\tilde{u}(x) = \bar{u}_i$ is

$$\tilde{u}(x) - u(x) = \underbrace{(\bar{u}_i - u(x_i))}_{\mathcal{O}(\Delta x^2)} - u_x(x_i)(x - x_i) + \mathcal{O}(\Delta x^2), \quad x \in I_i. \quad (39)$$

Since $|x - x_i| \leq \Delta x/2$, this error is $\mathcal{O}(\Delta x)$, and it is largest at the cell edges, where its size is about $\frac{1}{2}|u_x|\Delta x$. This is exactly where it hurts: the flux is evaluated at the interfaces. At $x_{i+1/2}$ Godunov's scheme uses the left state $\bar{u}_i$, while the true value there is $\bar{u}_i + \frac{1}{2}u_x\Delta x + \mathcal{O}(\Delta x^2)$. Every interface state, and therefore every flux, carries an $\mathcal{O}(\Delta x)$ error, and the scheme is first-order accurate. (Replacing the flat blocks by sloped lines reduces this interface error to $\mathcal{O}(\Delta x^2)$; this is the idea behind the second-order MUSCL scheme of Lecture 11.)

We now derive the modified equation explicitly, for linear advection with $a > 0$, where Godunov's scheme is the first-order upwind scheme (Section 4.3.1), written in the same cell-average notation $\bar{u}_i^n$ used throughout this lecture:

$$\bar{u}_i^{n+1} = \bar{u}_i^n - \nu (\bar{u}_i^n - \bar{u}_{i-1}^n), \quad \nu \equiv \frac{a\Delta t}{\Delta x}. \quad (40)$$

For a smooth solution $u(x, t)$, the cell average agrees with the point value at the cell center up to $\mathcal{O}(\Delta x^2)$: $\bar{u}_i^n = u(x_i, t^n) + \mathcal{O}(\Delta x^2)$. Since we are only tracking this modified equation up to $\mathcal{O}(\Delta x^2)$ ourselves, we may Taylor-expand $\bar{u}_i^n$ exactly as if it were the point value $u(x_i, t^n)$. Expanding every term about (x_i, t^n) :

$$\bar{u}_i^{n+1} = u + \Delta t u_t + \frac{\Delta t^2}{2} u_{tt} + \mathcal{O}(\Delta t^3), \quad \bar{u}_{i-1}^n = u - \Delta x u_x + \frac{\Delta x^2}{2} u_{xx} - \mathcal{O}(\Delta x^3), \quad (41)$$

where $u, u_t, u_{tt}, u_x, u_{xx}$ denote the smooth solution and its derivatives, all evaluated at (x_i, t^n) . Substituting into the scheme and using $\nu\Delta x = a\Delta t$:

$$\Delta t u_t + \frac{\Delta t^2}{2} u_{tt} + \mathcal{O}(\Delta t^3) = -\nu \left[\Delta x u_x - \frac{\Delta x^2}{2} u_{xx} + \mathcal{O}(\Delta x^3) \right] = -a\Delta t u_x + \frac{a\Delta t \Delta x}{2} u_{xx} + \mathcal{O}(\Delta x^3) \quad (42)$$

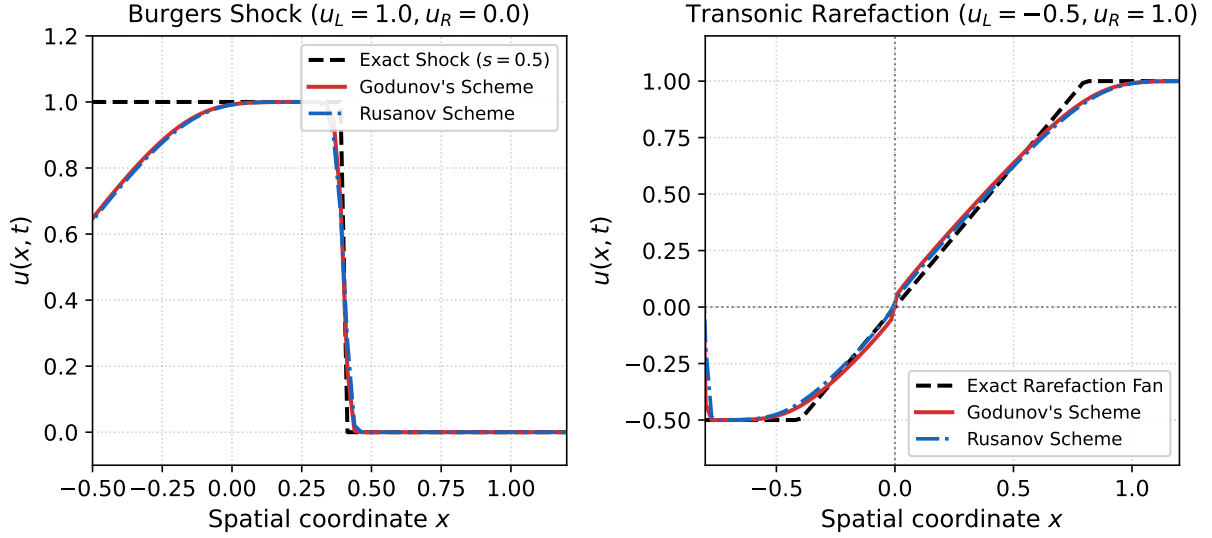


Figure 4: Numerical simulations with Godunov's scheme and Rusanov's scheme. Left: Moving Burgers shock ($u_L = 1.0, u_R = 0.0$); Godunov resolves the shock crisply and monotonically. Right: Transonic rarefaction ($u_L = -0.5, u_R = 1.0$); Godunov resolves the smooth expansion fan through the sonic point $u = 0$ without entropy violation.

Dividing through by Δt :

$$u_t + au_x = \frac{a\Delta x}{2}u_{xx} - \frac{\Delta t}{2}u_{tt} + \mathcal{O}(\Delta x^2) \quad (43)$$

This still contains the time-derivative term u_{tt} , which we eliminate using the PDE itself: to leading order, $u_t = -au_x$, so differentiating in x gives $u_{xt} = -au_{xx}$, and differentiating $u_t = -au_x$ in t instead gives $u_{tt} = -au_{xt} = a^2u_{xx}$ (up to higher-order corrections). Substituting this into equation (43):

$$u_t + au_x = \frac{a\Delta x}{2}u_{xx} - \frac{a^2\Delta t}{2}u_{xx} + \mathcal{O}(\Delta x^2) = \frac{a\Delta x}{2} \left(1 - \frac{a\Delta t}{\Delta x}\right) u_{xx} + \mathcal{O}(\Delta x^2) \quad (44)$$

which is exactly:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \underbrace{\frac{a\Delta x}{2}(1 - \nu)}_{\nu_{\text{num}}} \frac{\partial^2 u}{\partial x^2} + \mathcal{O}(\Delta x^2) \quad (45)$$

The numerical diffusion coefficient $\nu_{\text{num}} > 0$ (positive whenever $\nu = a\Delta t/\Delta x \leq 1$, i.e., exactly the CFL/monotonicity condition from Section 5.1) smears sharp discontinuities over several grid cells and rounds off smooth peaks. Reducing this diffusion, without losing the no-overshoot property, means reconstructing something better than a flat, constant value inside each cell (Step 1 above); we will develop exactly these higher-order reconstruction techniques in Lecture 11.

6 Approximate Riemann Solvers: The Rusanov (Local Lax-Friedrichs) Flux

Solving the exact nonlinear Riemann problem at every cell interface is straightforward for scalar equations, but for systems like the 3D compressible Euler equations, solving the exact iterative Riemann problem at every interface in a mesh of millions of cells is computationally prohibitive!

Modern CFD relies extensively on **Approximate Riemann Solvers** that capture the interface flux efficiently. The simplest and most universal approximate flux is the **Rusanov flux** (also called the **Local Lax-Friedrichs flux**, recalling the Lax-Friedrichs scheme from Lecture 7):

$$F_{i+1/2}^{\text{Rusanov}} = \frac{1}{2} (f(\bar{u}_i) + f(\bar{u}_{i+1})) - \frac{1}{2} s_{\max} (\bar{u}_{i+1} - \bar{u}_i) \quad (46)$$

where $s_{\max} = \max(|f'(\bar{u}_i)|, |f'(\bar{u}_{i+1})|)$ is the maximum local wave speed.

6.1 Physical Structure of the Rusanov Flux

The Rusanov flux decomposes into two intuitive components:

1. **Central Average:** $\frac{1}{2}(f(\bar{u}_i) + f(\bar{u}_{i+1}))$, which is second-order accurate in space but, exactly like the central-differenced FTCS scheme (Lecture 7), unconditionally unstable by itself.
2. **Scalar Dissipation:** $-\frac{1}{2}s_{\max}(\bar{u}_{i+1} - \bar{u}_i)$, which acts as an artificial viscosity (the same numerical-diffusion effect quantified in Section 5.3) proportional to the jump across the interface, stabilizing the scheme.

As shown in Figure 4, Rusanov is more dissipative than Godunov, smearing shocks over 4–5 cells instead of 2–3 cells, but it requires zero case-by-case logic and extends effortlessly to systems of equations!

7 Summary: Finite Difference vs. Finite Volume

Feature	Finite Difference Method (FDM)	Finite Volume Method (FVM)
Primary Unknown	Point value $u(x_j, t)$ at grid nodes	Cell average $\bar{u}_i(t) = \frac{1}{\Delta x} \int_{I_i} u \, dx$
Governing Equation	Differential form: $u_t + f_x = 0$	Integral form over each cell I_i
Behavior at Shocks	Taylor series breaks; derivatives blow up	Cell averages remain bounded and well-behaved
Discrete Conservation	Requires careful flux differencing	Guaranteed by construction via telescoping fluxes
Mesh Flexibility	Structured Cartesian or mapped grids	Arbitrary unstructured meshes (polygons, polyhedra)
Flux Evaluation	Differences of nodal point values	Exact or approximate Riemann solvers

8 Exercises

Exercises for Lecture 10

E1. Godunov Flux for Linear Advection with Negative Velocity: Let $u_t + au_x = 0$ with constant speed $a < 0$.

- (a) Solve the Riemann problem at interface $x = 0$ for $t > 0$.
- (b) Evaluate the exact interface state u^* and show that $F^{\text{Godunov}}(u_L, u_R) = au_R$.
- (c) Substitute this flux into the FVM update formula (8) and show that it yields the forward-differenced upwind scheme.
- (d) Determine the exact range of the CFL number $\nu = \frac{|a|\Delta t}{\Delta x}$ for which this scheme is monotone.

E2. Transonic Rarefaction Calculation: Consider Burgers' equation $u_t + (\frac{1}{2}u^2)_x = 0$ with cell averages $\bar{u}_1 = -2.0$ and $\bar{u}_2 = +3.0$ across interface $x_{3/2}$.

- (a) Is the Riemann solution a shock or a rarefaction fan? Justify your answer using characteristic speeds.
- (b) Determine the propagation speeds of the left edge and the right edge of the wave.
- (c) Find the exact state u^* along the interface line $x/t = 0$.
- (d) Compute the Godunov flux $F_{3/2}^{\text{Godunov}}$ and compare it to the Rusanov flux $F_{3/2}^{\text{Rusanov}}$. Explain why the Rusanov flux is non-zero while the Godunov flux is identically zero.

E3. Monotonicity of the Rusanov Scheme for Scalar Conservation Laws: Consider the general scalar conservation law $u_t + f(u)_x = 0$ discretized using the Finite Volume Method with the Rusanov flux (46).

- (a) Write the full update operator $\bar{u}_i^{n+1} = \mathcal{H}(\bar{u}_{i-1}^n, \bar{u}_i^n, \bar{u}_{i+1}^n)$ assuming a constant maximum wave speed $s_{\max} = \alpha > 0$.
- (b) Compute the partial derivatives $\frac{\partial \mathcal{H}}{\partial \bar{u}_{i-1}}$, $\frac{\partial \mathcal{H}}{\partial \bar{u}_i}$, and $\frac{\partial \mathcal{H}}{\partial \bar{u}_{i+1}}$.
- (c) Prove that the scheme is monotone provided that the CFL condition $\alpha \frac{\Delta t}{\Delta x} \leq 1$ is satisfied.

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